

1. Write a Pandas program to select the distinct department id from employee's file.**Aim:**

To write a Pandas program to select the distinct department IDs from the employee's file.

Program:

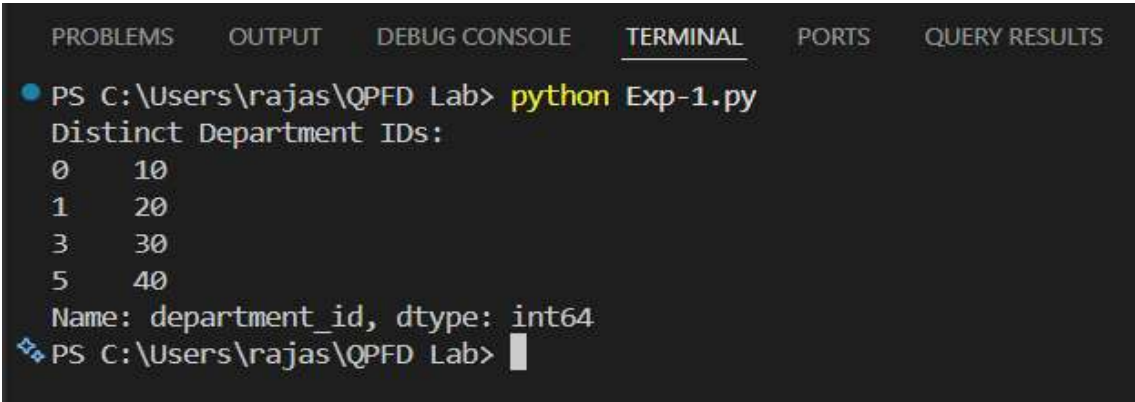
```
import pandas as pd
```

```
employees = pd.read_csv("employees.csv")
```

```
distinct_dept = employees["department_id"].drop_duplicates()
```

```
print("Distinct Department IDs:")
```

```
print(distinct_dept)
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  QUERY RESULTS

PS C:\Users\rajas\QPFD Lab> python Exp-1.py
Distinct Department IDs:
0      10
1      20
3      30
5      40
Name: department_id, dtype: int64
PS C:\Users\rajas\QPFD Lab>
```

Result:

The program successfully selected and displayed the distinct department IDs from the employees file.

2. Write a Pandas program to display the IDs of employees who have done two or more**Aim:**

To write a Pandas program to display the IDs of employees who have done two or more jobs.

Program:

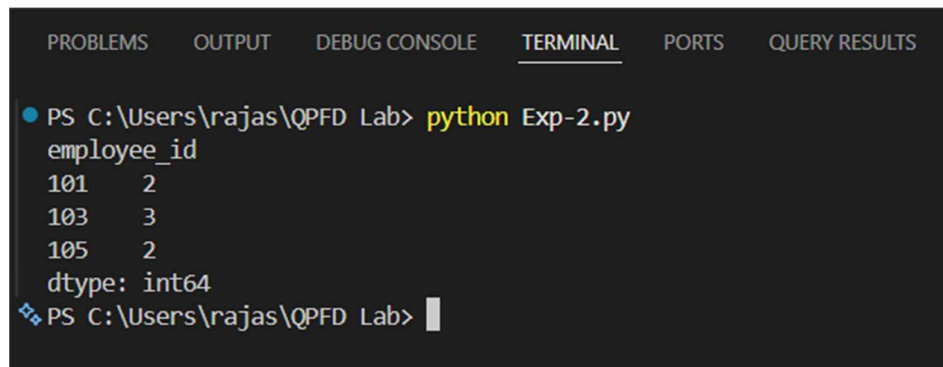
```
import pandas as pd
```

```
jobs = pd.read_csv("employee_jobs.csv")
```

```
job_count = jobs.groupby("employee_id").size()
```

```
result = job_count[job_count >= 2]
```

```
print(result)
```

Output:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS
PS C:\Users\rajas\QPFD Lab> python Exp-2.py
employee_id
101      2
103      3
105      2
dtype: int64
PS C:\Users\rajas\QPFD Lab>
```

Result:

The program successfully displayed the IDs of employees who have done two or more jobs.

3. Write a Pandas program to display the details of jobs in descending sequence on job title.**Aim:**

To write a Pandas program to display the details of jobs in descending order based on the job title.

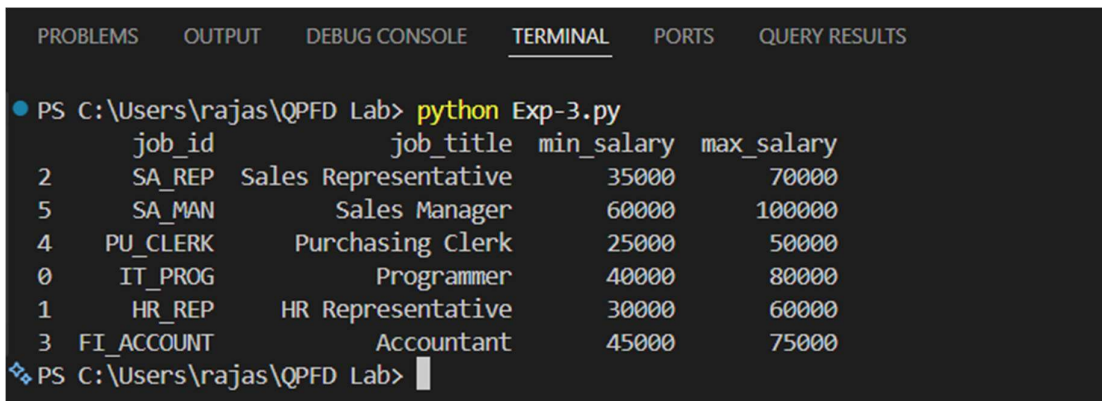
Program:

```
import pandas as pd
```

```
jobs = pd.read_csv("jobs.csv")
```

```
result = jobs.sort_values(by="job_title", ascending=False)
```

```
print(result)
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  QUERY RESULTS
```

```
● PS C:\Users\rajas\QPFD Lab> python Exp-3.py
```

	job_id	job_title	min_salary	max_salary
2	SA_REP	Sales Representative	35000	70000
5	SA_MAN	Sales Manager	60000	100000
4	PU_CLERK	Purchasing Clerk	25000	50000
0	IT_PROG	Programmer	40000	80000
1	HR_REP	HR Representative	30000	60000
3	FI_ACCOUNT	Accountant	45000	75000

```
❖ PS C:\Users\rajas\QPFD Lab> █
```

Result:

The program successfully displayed the job details in descending order based on the job_title column.

4. Write a Pandas program to create a line plot of the historical stock prices of Alphabet Inc. between two specific dates.

Aim:

To write a Pandas program to create a line plot of the historical stock prices of Alphabet Inc. between two specific dates.

Program:

```
import pandas as pd
import matplotlib.pyplot as plt

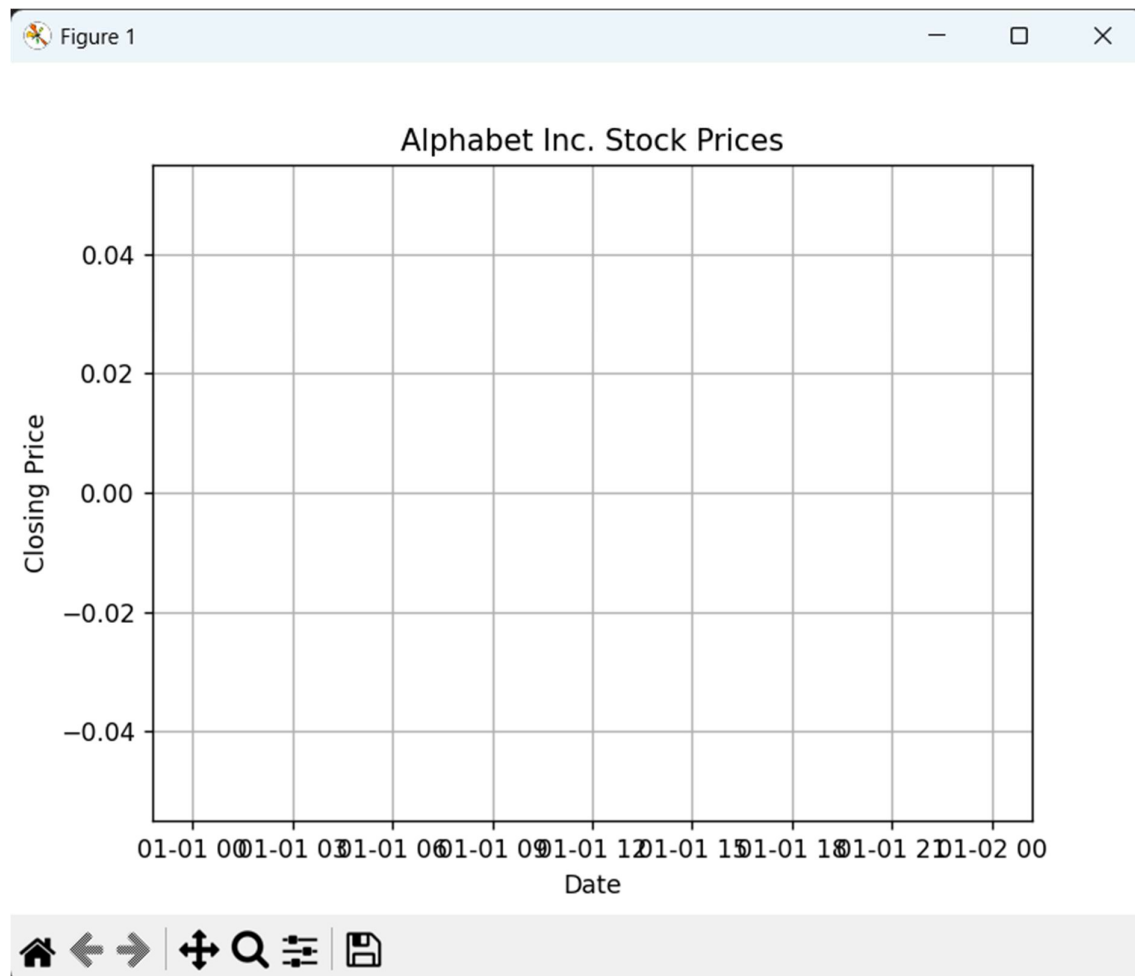
data = pd.read_csv("alphabet_stock.csv")

data["Date"] = pd.to_datetime(data["Date"])

filtered = data[(data["Date"] >= "2024-01-02") & (data["Date"] <= "2024-01-06")]

plt.plot(filtered["Date"], filtered["Close"], marker="o")
plt.title("Alphabet Inc. Stock Prices")
plt.xlabel("Date")
plt.ylabel("Closing Price")
plt.grid(True)
plt.show()
```

Output:



Result:

The program successfully created a line plot of Alphabet Inc.'s historical stock prices between the specified dates.

5. Write a Pandas program to create a bar plot of the trading volume of Alphabet Inc. stock between two specific dates.

Aim:

To write a Pandas program to create a bar plot of the trading volume of Alphabet Inc. stock between two specific dates.

Program:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
data = pd.read_csv("alphabet_stock.csv")
```

```
data["Date"] = pd.to_datetime(data["Date"])
```

```
filtered = data[(data["Date"] >= "2024-01-02") & (data["Date"] <= "2024-01-06")]
```

```
plt.bar(filtered["Date"], filtered["Volume"])
```

```
plt.title("Alphabet Inc. Trading Volume")
```

```
plt.xlabel("Date")
```

```
plt.ylabel("Trading Volume")
```

```
plt.xticks(rotation=45)
```

```
plt.show()
```

Output:



Result:

The program successfully created a bar plot of the trading volume of Alphabet Inc. stock between the specified dates.

6. Write a Pandas program to create a scatter plot of the trading volume/stock prices of Alphabet Inc. stock between two specific dates.**Aim:**

To write a Pandas program to create a scatter plot of the trading volume and stock prices of Alphabet Inc. stock between two specific dates.

Program:

```
import pandas as pd
import matplotlib.pyplot as plt

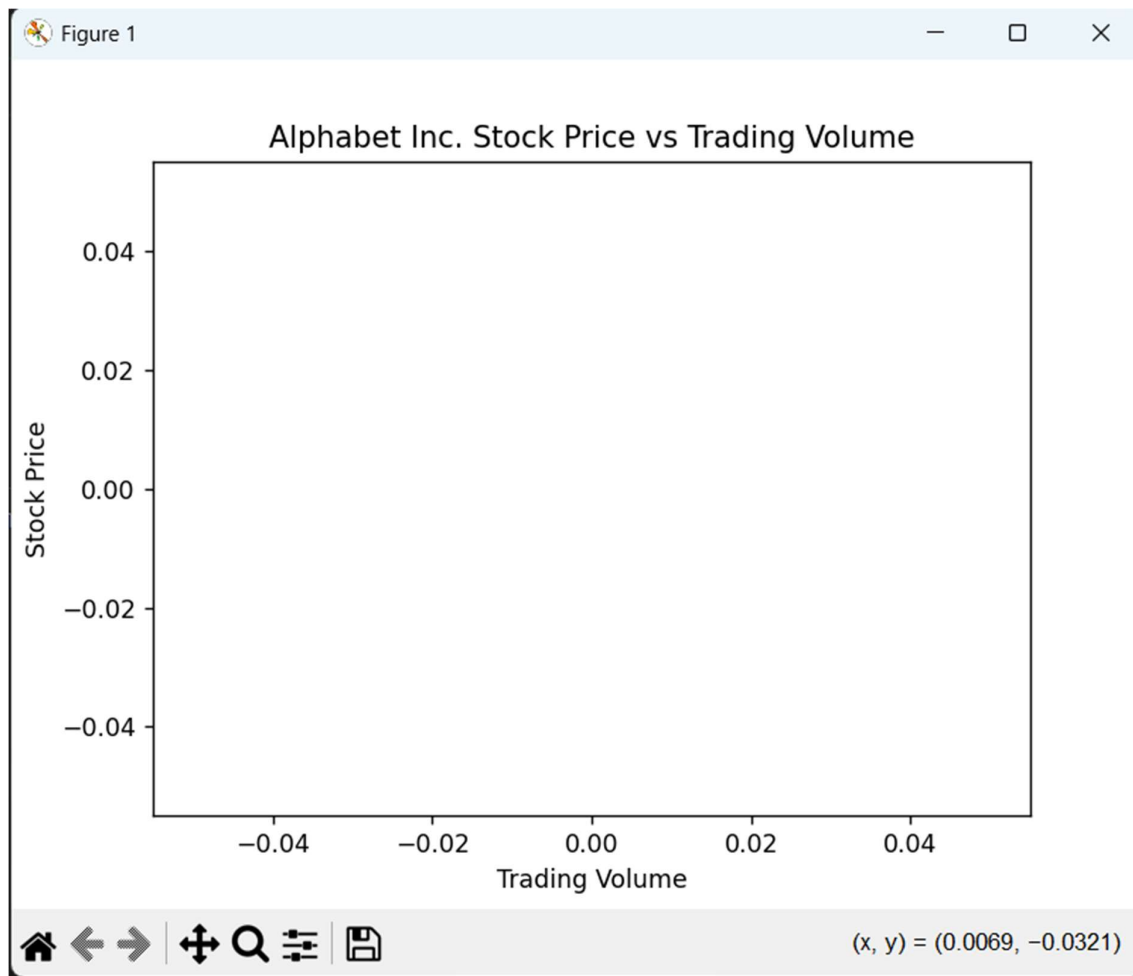
data = pd.read_csv("alphabet_stock.csv")

data["Date"] = pd.to_datetime(data["Date"])

filtered = data[(data["Date"] >= "2024-01-02") & (data["Date"] <= "2024-01-06")]

plt.scatter(filtered["Volume"], filtered["Close"])
plt.title("Alphabet Inc. Stock Price vs Trading Volume")
plt.xlabel("Trading Volume")
plt.ylabel("Stock Price")
plt.show()
```

Output:



Result:

The program successfully created a scatter plot of the trading volume and stock prices of Alphabet Inc. stock between the specified dates.

